

Gabrielle Johnson's Co-Design Canvas — HCI: Nurse Triage Interface

1. Problem

The workflow- Mercer General sees 70–95 patients on a normal day and up to 170 on a heavy one. Every patient is triaged by a nurse into ESI 1–5 in a 3–5 minute assessment. That level sets the target time-to-be-seen: immediate, ≤ 10 min, ≤ 30 min, ≤ 60 min, or can wait. The decision is made from front-door information alone like vitals, chief complaint, arrival mode etc.

Currently Mercer faces a 20% ESI disagreement amongst nurses. An ESI 2 patient has a 10-minute target; an ESI 3 patient has 30 minutes and, on a heavy day, considerably longer. Hence its crucial to prevent undertriage and alert staff when patients are deteriorating in the waiting room.

Users.

	Group	Characteristics	Needs
Primary	Triage nurses	Mobile;interrupt ed constantly; 3–5 minutes per patient;	Trustworthy ESI levels for each patient; a log of each patient that needs re-triaging
Secondary	Charge nurse	One per shift, owns flow and zone assignment	Visibility of patients who are waiting, unreviewed, or deteriorating

Goals.

	Short-term	Long-term
Nurses	A defensible second opinion inside the existing 3–5 minutes	Fewer emergent patients placed in lower categories
Charge nurse	Ensure patients flow is efficient and safe	Reducing esi level disagreements to less than 5% and improving retriaging

		efficiency, reducing interruptions from patients
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What the system does not do. It does not assign the final ESI level, hence the nurses' judgement is still crucial. ESI 1 is out of scope as they tend to be clinically apparent and admitted immediately.

2. Ethics

Area	View	Problem	Solution
Automation bias	User	A nurse may defer exercising their judgement and wait on the model's response.	Request the nurse's own assessment; declining requires a reason; agreement rates are monitored by the charge nurse.
	System	The model cannot express its own limitations	Confidence and data-quality flags rendered on-screen; contradicting evidence displayed alongside supporting evidence
Screen truncation	User	A phone sized display fits two or three features before scrolling, making acceptance easier than scrutiny	Contradicting features are shown above the scroll; the accept button sits after the reasoning, so the nurse must scroll past the arguments to agree
	System	Ranking features by magnitude will always surface the same	Reasoning is capped at three supporting and two contradicting

		few, hiding rare but decisive signals	features, chosen for decision relevance rather than raw magnitude
Alert fatigue	User	The chosen configuration flags roughly 1,988 patients as ESI 2 who are not	Prompts raised only for downward disagreements, never upward; alert volume bounded per shift and reviewed with staff
	System	An interface that interrupts on every disagreement trains staff to dismiss it	Disagreement is displayed passively; only deterioration during the wait interrupts
Accountability	User	Override reason logs are also a record of individual nurses' decisions	Restricted in policy to model retraining and audit, agreed with staff before deployment; the nurse remains decision maker of record

3. Design Guidelines

Advantage guidelines- The system should focus on what software does better than humans under time constraints: applying rules consistently, retaining information and considering many weak signals at once. Judgement, reassurance etc. must remain with the nurse.

Ethical guidelines- Every screen should make disagreeing as easy as agreeing. Where the model is known to be weak, it is displayed.

Environment guidelines- This is a strict clinical process under severe time pressure, so optimise for efficiency and safety, not exploration. Hence, the interface must survive interruptions, should be easy on the eyes and easy to navigate.

Interaction guidelines- visual display, tap input.

Behaviour guidelines. Simple and stateless. The system has no internal drivers, no goals of its own, and no memory of this nurse’s previous decisions that could influence the current one. It reacts to input, presents, and waits

Form guidelines. The “form” of a screen tool should look similar to existing Mercer triage forms

4. MVP

Where and when. Anywhere in the department, on a carried handheld device, 24/7

Role. Colleague- A second opinion

Personality. None- Formal/Clinical,neutral language throughout.

Connection to systems. The CatBoost model and its SHAP explainer; the queue and assignment engine;. There is no electronic patient record at Mercer to integrate with, so the interface must stand alone

Interaction modalities.

	Modalities used
Input	Touch (screen taps), device stream (vitals from the patient’s watch via QR pairing), text entry (chief complaint)
Output	Screens (primary), haptics

Interaction flow — the ESI decision.

	Before	During	After
Nurse	Scans the patient’s watch QR to pair the record	Accepts or rejects each captured vital; takes and enters chief complaints; reads the suggestion and its reasoning	Accepts or declines with a reason; writes the final level to the paper record
System	Receives paired vitals; checks age and completeness	Withholds the suggestion until chief complaints is entered; then	Logs the decision, the reason for any override, and the

		renders level, supporting and contradicting evidence, quality flags, blind-spot card if applicable	elapsed time; returns the patient to the acuity-ordered queue
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5. Environment

Where	Wherever the nurse is — triage bay, cubicle, corridor, fast-track area. Bright overhead lighting, high ambient noise, gloved hands, often standing. The physical setting is not fixed
When	24/7 , Load rises after long weekends and through tourist high season
Users	Triage nurses
Secondary users	Charge nurse, who receives escalations for unreviewed and overdue patients
Simultaneous users	Few — several nurses may hold queues at once, each seeing only his/her own caseload. Assignment is weighted by acuity rather than headcount, since two ESI 2 patients are more work than five ESI 4s
Data collection	Patient id, user id , Vitals accepted or rejected, chief complaint, age, arrival mode, the suggested level, the final level, and the reason for any disagreement.
External sensors and actuators	The vitals arrive from the patient's watch. Output is screen only; the handheld's vibration is reserved for deterioration alerts so it is never confused with routine traffic
Connection to systems	No hospital IT dependency; must tolerate network loss without losing an in-progress assessment, and fall back to conventional triage when unavailable
